

# Gen2 Workspace — Port Consolidation Report

Change record: all application-level network ports moved into a single 8100-8199 series

## 1. Summary

All 14 previously-scattered ports have been consolidated into a single 8100-8199 range, grouped in blocks of 10 by subsystem, resolving the prior three-way collision on port 8080. 15 edits were made across 15 files (some ports appear in more than one file that must stay in sync). Every change was verified by re-parsing the file and, where a colcon package exists, rebuilding it.

Action needed: any of these services already running were started with the OLD port bound and won't pick up the new one until restarted. Bookmarks/scripts pointing at the old `http://host:PORT` URLs also need updating to the new ones in the table below.

## 2. Migration Map (old → new)

| Old        |   | New  | Service / file                                                                                                                                                                                                                                                                      |
|------------|---|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5001       | → | 8100 | automation - Bringup Launch Control (Flask)<br>automation/automation/launch_control_app.py                                                                                                                                                                                          |
| 15000      | → | 8101 | gen2 ROS-TCP manager (Flask)<br>gen2/gen2/ros_tcp_manager.py                                                                                                                                                                                                                        |
| 25000      | → | 8102 | gen2 leader controller UI (Flask)<br>gen2/gen2/leader_controller_ui.py                                                                                                                                                                                                              |
| 10000      | → | 8103 | ROS-TCP-Endpoint (Unity bridge) - server default,<br>launch/endpoint.py param, and ros_tcp_manager.py's<br>free_port() call, all updated together<br>tele/ROS-TCP-Endpoint/ros_tcp_endpoint/server.py,<br>tele/ROS-TCP-Endpoint/launch/endpoint.py,<br>gen2/gen2/ros_tcp_manager.py |
| 5005 (UDP) | → | 8104 | Raw leader-controller socket listener<br>tele/gen2_leader/gen2_leader/Gen2Leader_raw.py                                                                                                                                                                                             |
| 5000       | → | 8110 | gesture_management - Teach & Replay Console (Flask)<br>gesture_management/gesture_management/gesture_management_a<br>pp.py                                                                                                                                                          |

|                                 |   |             |                                                                                                                                                                                                |
|---------------------------------|---|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7000                            | → | <b>8120</b> | Daksha Dashboard (Flask) - default AND<br>project.config.yaml's daksha_ui_port/daksha_ui_url<br>ui/Clients_UI/ihub_ui/Daksha/app.py,<br>ui/Clients_UI/project.config.yaml                      |
| 7001                            | → | <b>8121</b> | Daksha sub-UI (Flask)<br>ui/Clients_UI/ihub_ui/Daksha/ui/app.py                                                                                                                                |
| 70002 (typo in<br>source)       | → | <b>8122</b> | Daksha Viveka camera-stream UI - also fixed a stray extra<br>digit in the old hardcoded default<br>ui/Clients_UI/ihub_ui/Daksha/ui/viveka_camera_ui.py                                         |
| 8123 (default)                  | → | <b>8130</b> | joint_analyzer server<br>ui/analyzer/joint_analyzer/server.py                                                                                                                                  |
| 8200 (default)                  | → | <b>8131</b> | joint_analyzer plot server<br>ui/analyzer/joint_analyzer/plot_server.py                                                                                                                        |
| 8080                            | → | <b>8140</b> | Data collection management web UI - the<br>data_collection_management copy (config-driven, newer)<br>ui/dc/data_collection_management/data_collection_managemen<br>t/web_data_management_ui.py |
| 8080                            | → | <b>8141</b> | Daksha API bridge camera streamer<br>global/daksha_api_bridge_leader/daksha_api_bridge_leader/c<br>amera_streamer.py                                                                           |
| 8080 (unused config →<br>value) | → | <b>8142</b> | Data Collection Terminal - project.config.yaml's<br>data_collection_port/data_collection_url<br>ui/Clients_UI/project.config.yaml                                                              |
| 8080                            | → | <b>8143</b> | Data collection management web UI - the ros_pkg copy<br>(older, hardcodes domain_id=3)<br>ui/Clients_UI/final/src/ros_pkg/ros_pkg/web_data_managemen<br>t_ui.py                                |

### 3. Verification Performed

- Every edited Python file (15 files) re-parsed cleanly with ast.parse - no syntax errors introduced by the renumbering.
- project.config.yaml re-loaded cleanly with yaml.safe\_load after editing the four network values it owns.

- `colcon build --packages-select automation gesture_management gen2 gen2_leader data_collection_management daksha_api_bridge_leader ros_tcp_endpoint` - all 7 ROS packages built successfully.
- `ui/Clients_UI/ihub_ui/Daksha` and `ui/analyzer/joint_analyzer` are plain script directories (no `package.xml`) run directly with `python3`, not `colcon` packages - verified by syntax check only, which is all that applies to them.

Result: all edits applied cleanly. No syntax errors, no failed builds.

## 4. Notes & Follow-ups

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- The two data-collection web UIs (`ui/dc/data_collection_management` and `ui/Clients_UI/final/.../ros_pkg`) are NOT identical files (2613 vs 3035 lines, different default `ROS_DOMAIN_ID`) - they were kept on separate ports (8140, 8143) rather than assuming one is dead code and collapsing them onto one port. Worth confirming which one is actually in active use and retiring the other, but that's a code decision, not a port decision.
  - `project.config.yaml`'s `data_collection_port` (now 8142) is not actually read by either data-collection script - both hardcode their own port literal. The value has been kept in sync and given its own reserved slot in the new range, but wiring it up so the config is a real source of truth (like it already is for `daksha_ui_port`) is a follow-up, not done here.
  - The ROS-TCP-Endpoint's port was declared in three places that all have to agree (`server.py`'s own default, `launch/endpoint.py`'s launch parameter, and `ros_tcp_manager.py`'s `free_port()` pre-flight kill) - all three were updated together to 8103.
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Generated after applying the port renumbering proposed in the prior port-inventory report. New allocation: 8100-8109 bringup/teleop control, 8110-8119 teach/record consoles, 8120-8129 Daksha UI stack, 8130-8139 analyzer tools, 8140-8149 data collection.